Agent-Based Infrastructures for Diachronic Digital Humanities

Cohort

Agents for retrieving, comparing and documenting textual evidence

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Acknowledgement

Thank you, Michael Radich

Texts

A modified, selected CBETA-derived corpus

Catalogue

Translator labels and mixed historical groups

Vocabulary

Curated strings for Dharmarakṣa research

Cohort builds the profiles and calculates the rankings.

Why we built Cohort

Let researchers direct agents and inspect the evidence behind their proposals.

Today's presentation

Use Buddhist texts to show the functions,
where results can mislead, and what the researcher decides.

We are demonstrating research tools, not reporting a new translator attribution.

Agent support for diachronic research

Study texts and their relationships across time.

Corpus retrieval

Find relevant texts and passages.

Semantic analysis

Find passages with related content.

Alignment

Locate matching wording and positions.

Interpretation

Propose explanations and alternatives.

Provenance Keep sources and checks attached to proposals.

Researcher decisions Inspect the evidence; accept or reject proposals.

Functions

Corpus	Find passages and read them in context.
Vocabulary comparison	Compare a text’s wording with texts labelled by translator.
Inquiry	Give agents a research question and inspect their actions.
Findings	Read the agents’ proposals, citations and checks.
Graph	Follow links between sources, proposals and decisions.

Example research questions

Two Lotus translations

Dharmarakṣa 竺法護 · T0263
Kumārajīva 鳩摩羅什 · T0262

Can we find related passages when the wording differs?

Text and later commentary

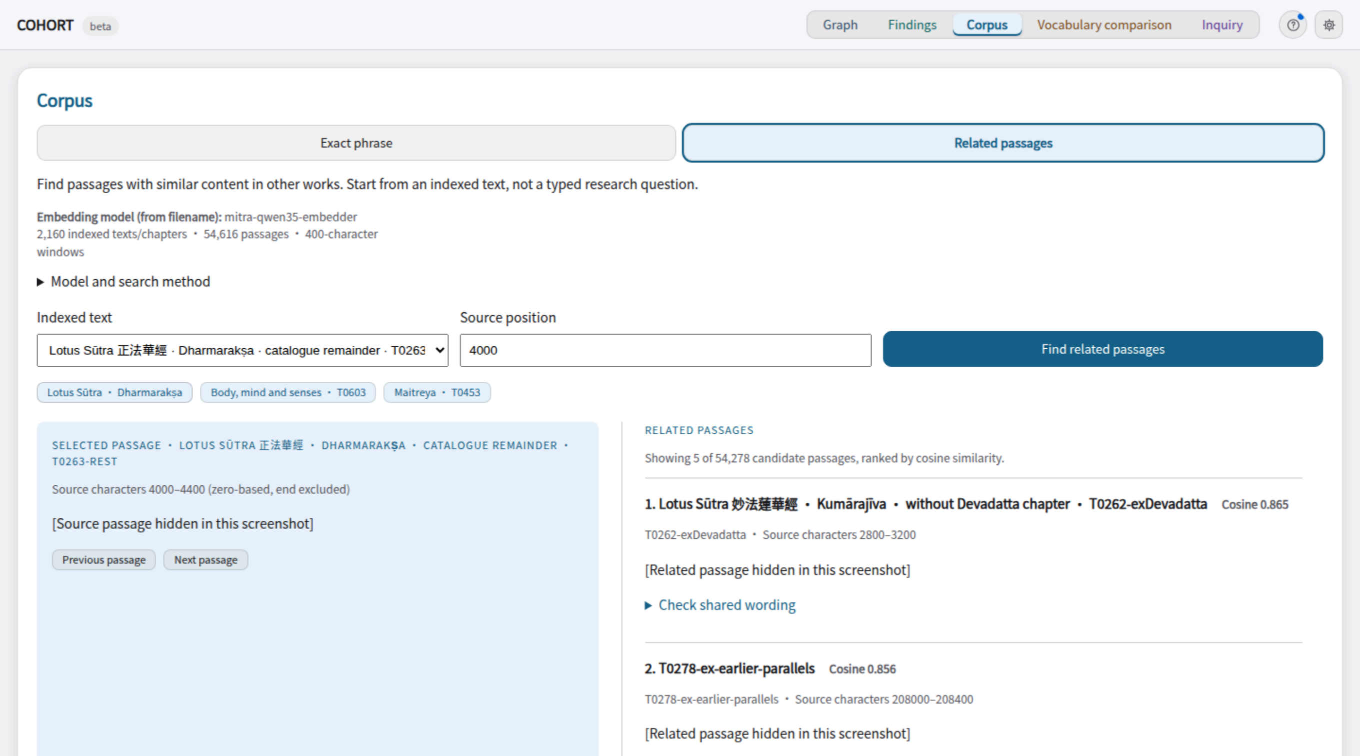
Yin chi ru jing 陰持入經 · T0603
Body, mind and senses

Could shared wording mislead us about who translated this text?

Known relationships let us inspect what the tools get right and wrong.

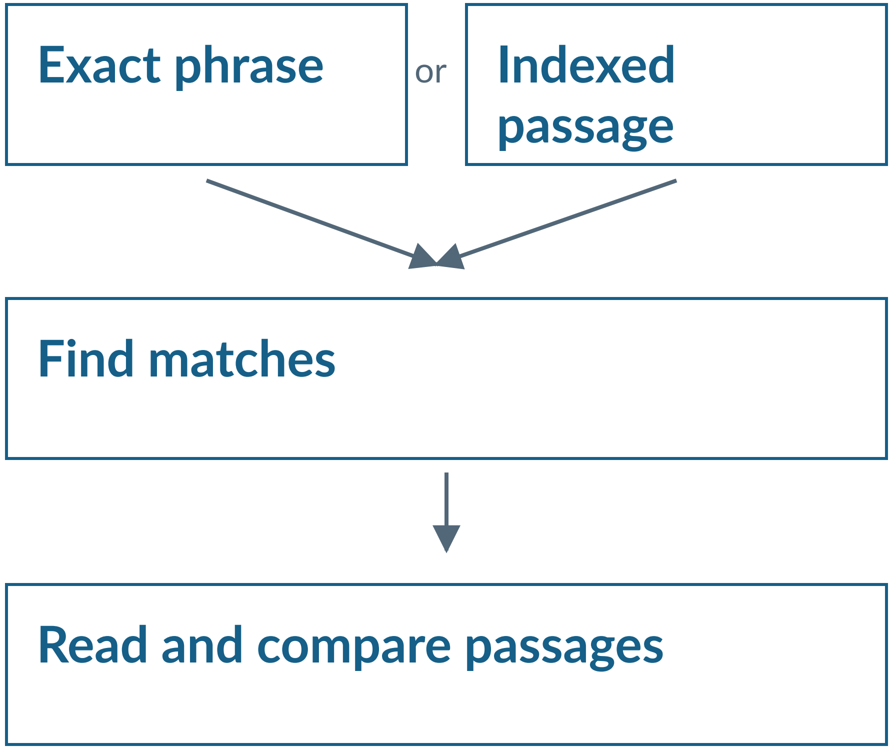
Find passages and read them in context.

Interface



Exact search or stored-vector retrieval.

How to use it



Browsing does not add graph records.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How related passages are found

mitra-qwen35-embedder

Selected passage

Use its stored vector

→

Cosine similarity

Compare with other works

→

Related passages

Rank candidates to read

Corpus → Related passages; agents can call semantic_neighbors.

Model name comes from the index filename; its exact revision is unrecorded.

Example: related content, little shared wording

Earlier Lotus translation

Dharmarakṣa · T0263-rest
Source position 4000

cosine 0.865 →

Later Lotus translation

Kumārajīva · T0262-exDevadatta
Source position 2800

Longest exact shared sequence: 3 Chinese characters

Shared wording is checked separately, with positions in both sources.

What vocabulary comparison calculates

Vocabulary list

Choose strings to count →

Target text

Count those strings

↓

Group profiles

Texts pooled by catalogue label

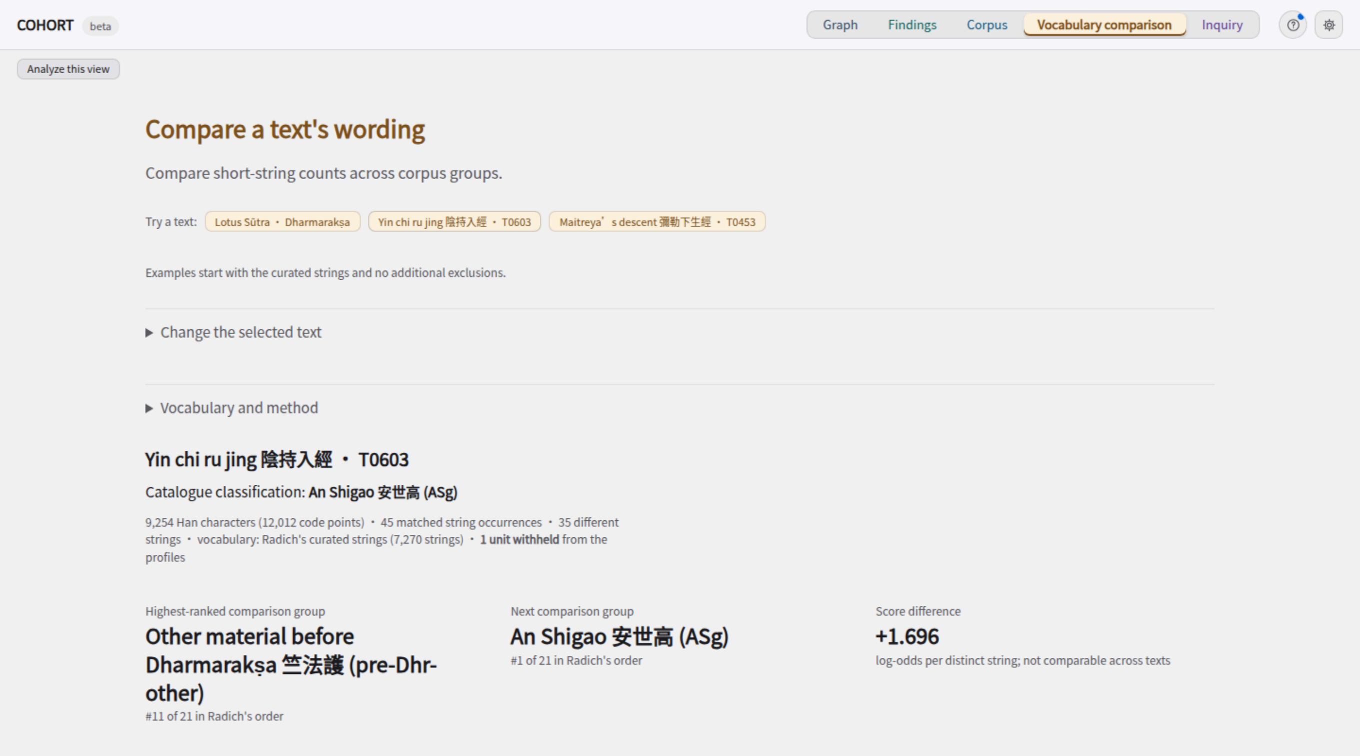
↓

Compare usage → rank the catalogue groups

13 usable groups in this dataset. The target's whole work is withheld.

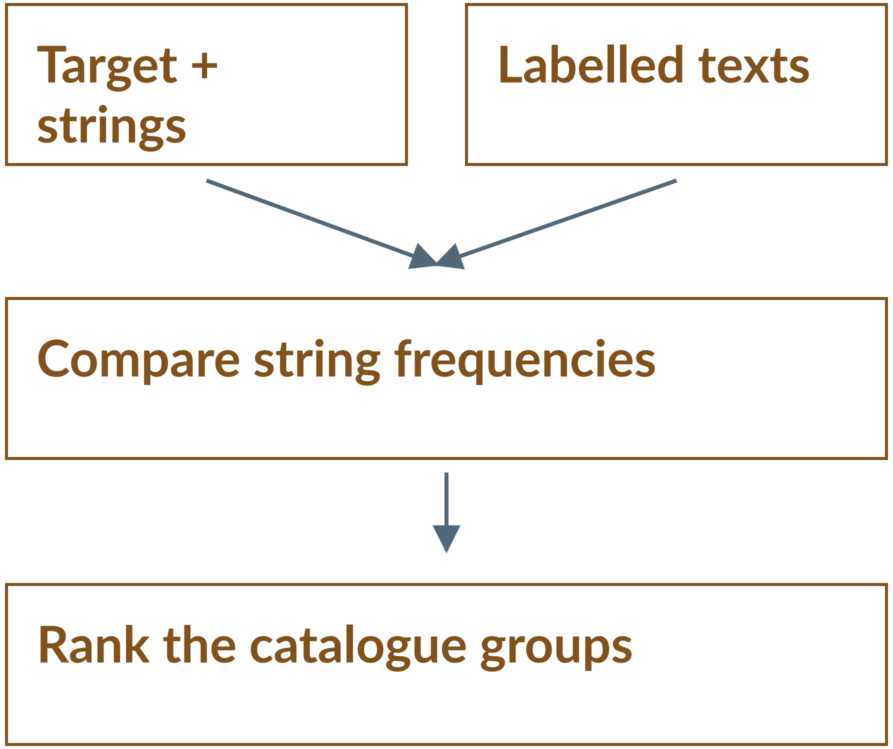
Compare a text’s wording with texts labelled by translator.

Interface



Exclude a comparison text and repeat.

How to use it



The vocabulary selects which strings count.

Screenshot: source passages hidden where applicable. Diagram: schematic.

T0603: remove the later commentary

With T1694

First: Other material before Dharmarakṣa 竺法護

Margin over next group: 1.696

→

Without T1694

First: An Shigao 安世高

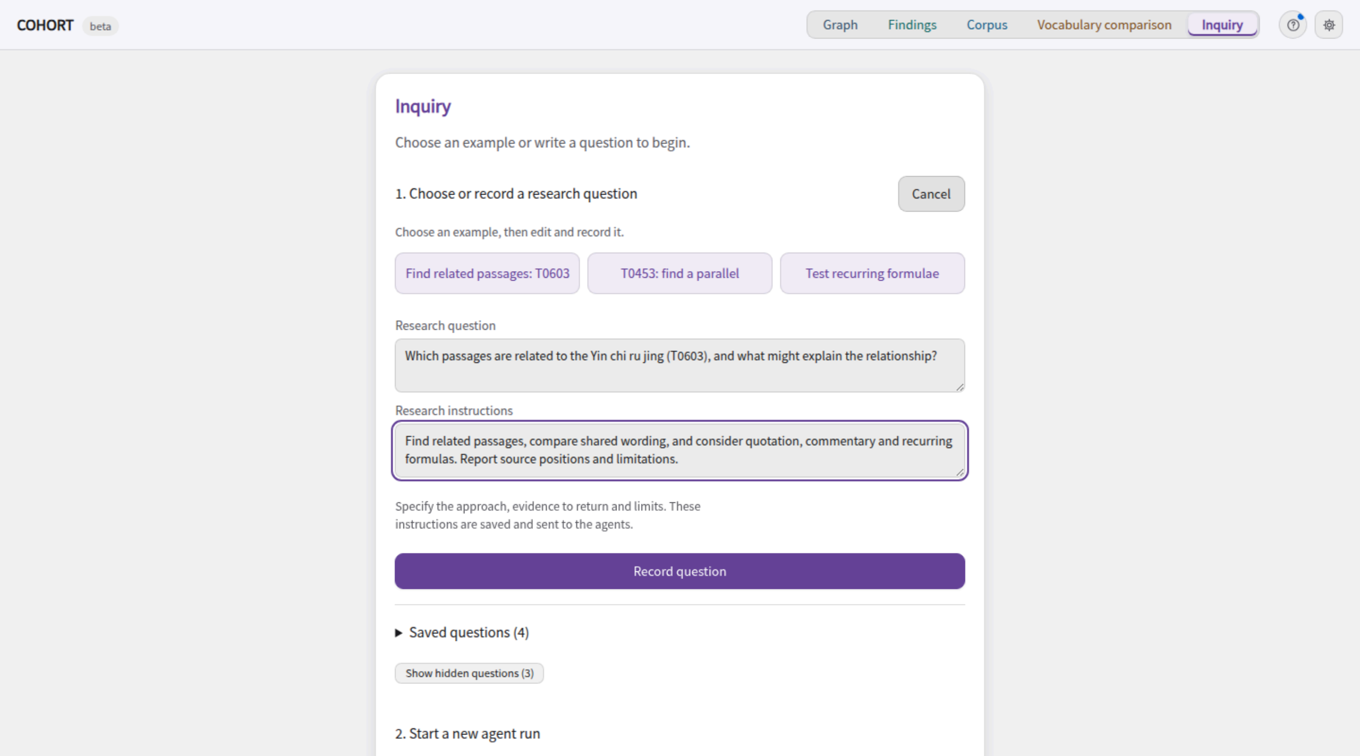
Margin over next group: 0.096

T1694 is the commentary on T0603. Its wording affects the ranking.

This agrees with the accepted attribution after exclusion; it does not establish one.

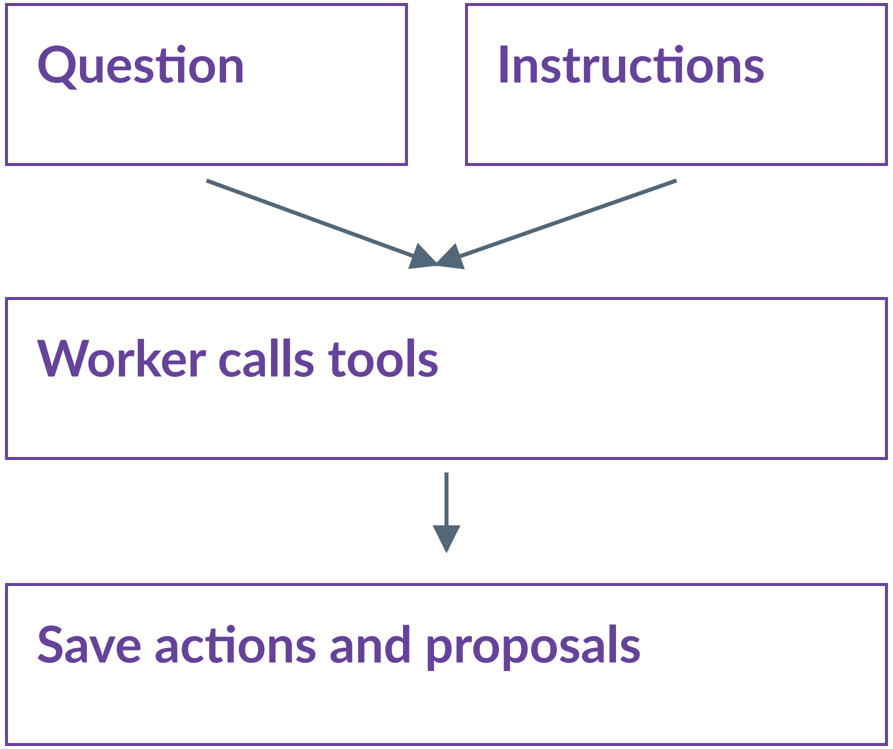
Give agents a research question and inspect their actions.

Interface



A separate reviewer records checks.

How to use it



Reopen completed work in Run results.

Screenshot: source passages hidden where applicable. Diagram: schematic.

From passages to research proposals

Agents can compare material, suggest explanations and propose follow-up searches.

Read and compare

Retrieve passages
Summarise observations

→

Propose

Possible explanations
Questions to investigate

→

Inspect and decide

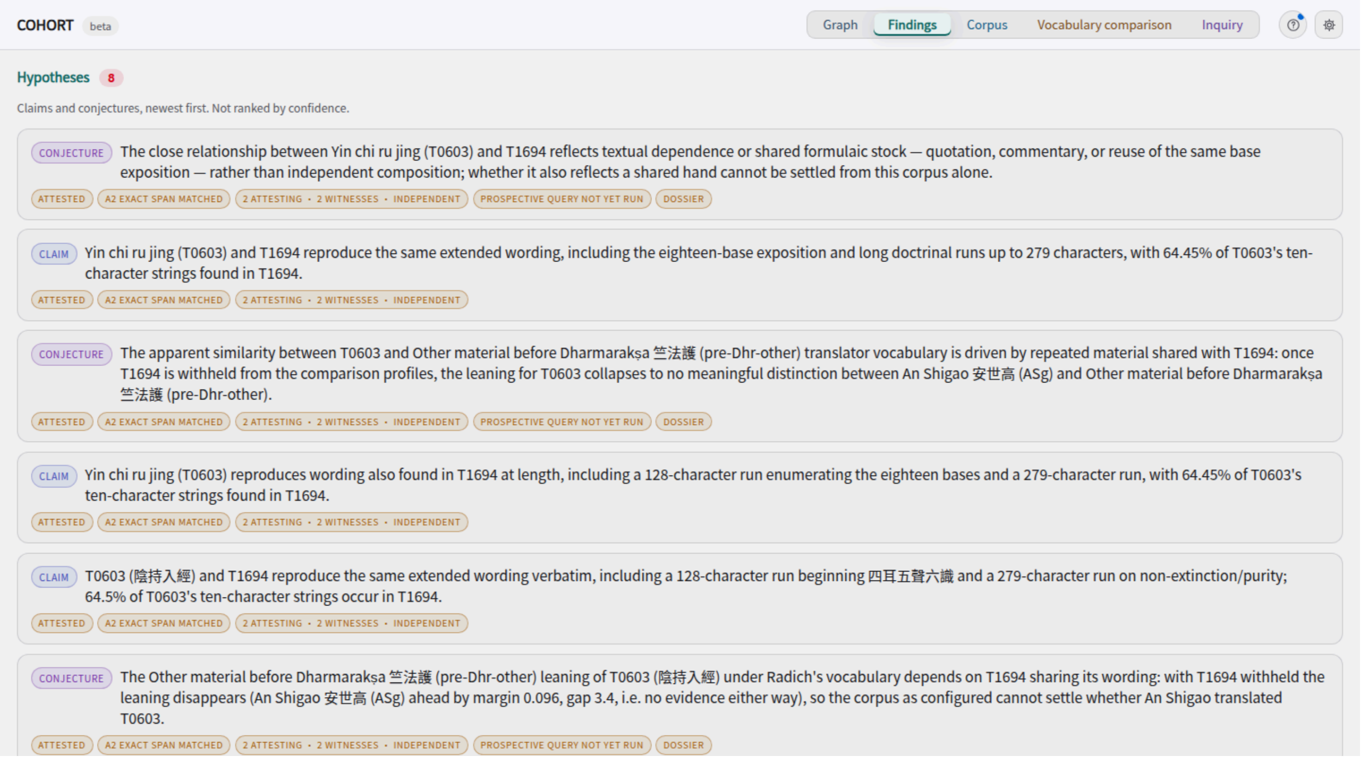
Read the cited sources
Choose what to pursue

Graph Keeps proposals linked to passages, checks and researcher decisions.

A citation that resolves can still support a weak interpretation.

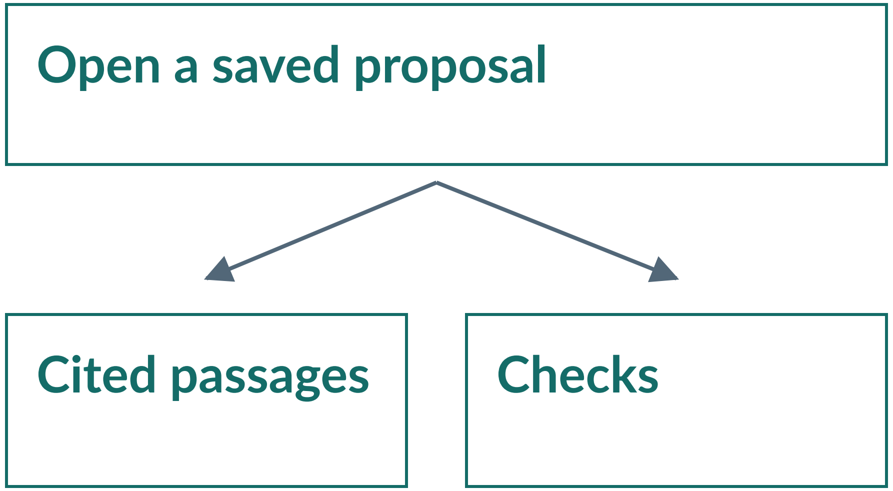
Read the agents’ proposals, citations and checks.

Interface



Follow a citation into Graph to inspect its source.

How to use it



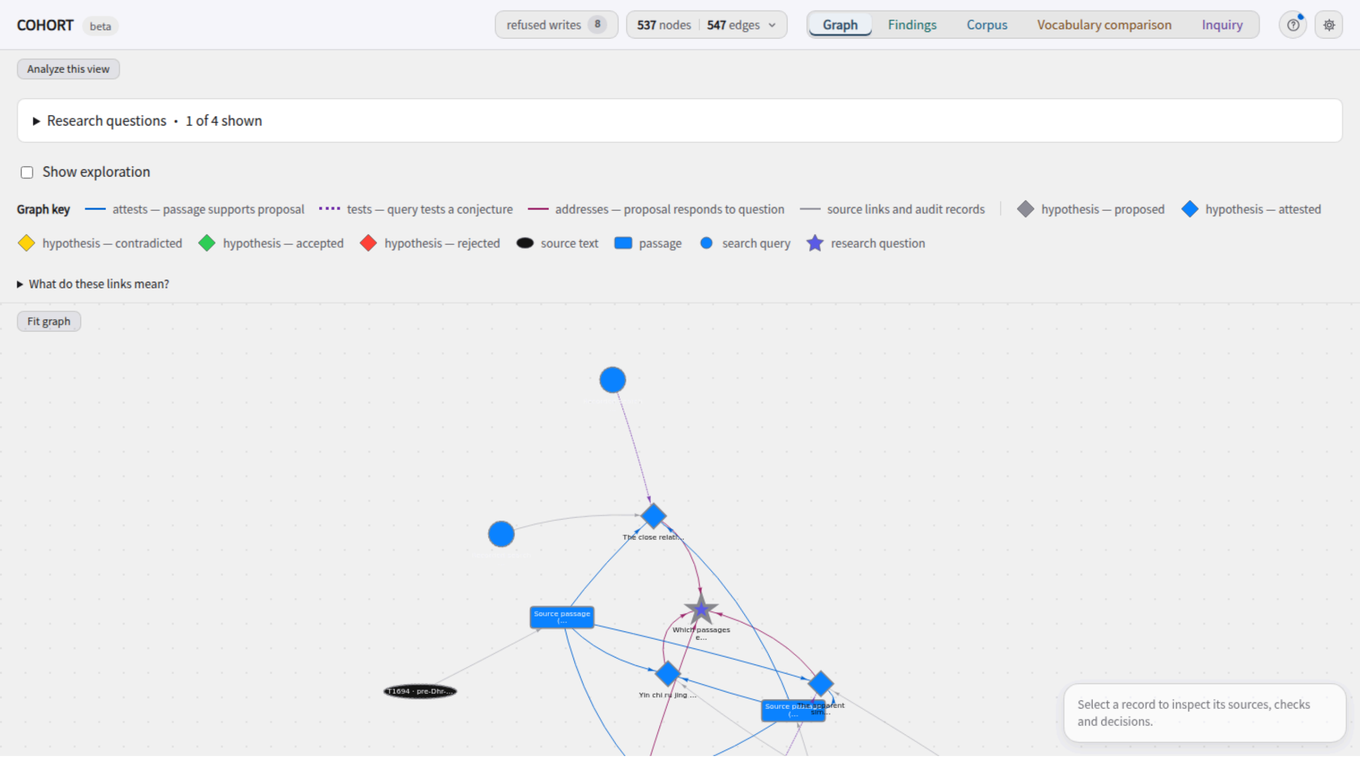
Read its explanation and alternatives.

Inspect one proposal at a time.

Screenshot: source passages hidden where applicable. Diagram: schematic.

Follow links between sources, proposals and decisions.

Interface



Show exploration adds recorded tool activity.

How to use it

Select a research question

↓

Inspect linked records

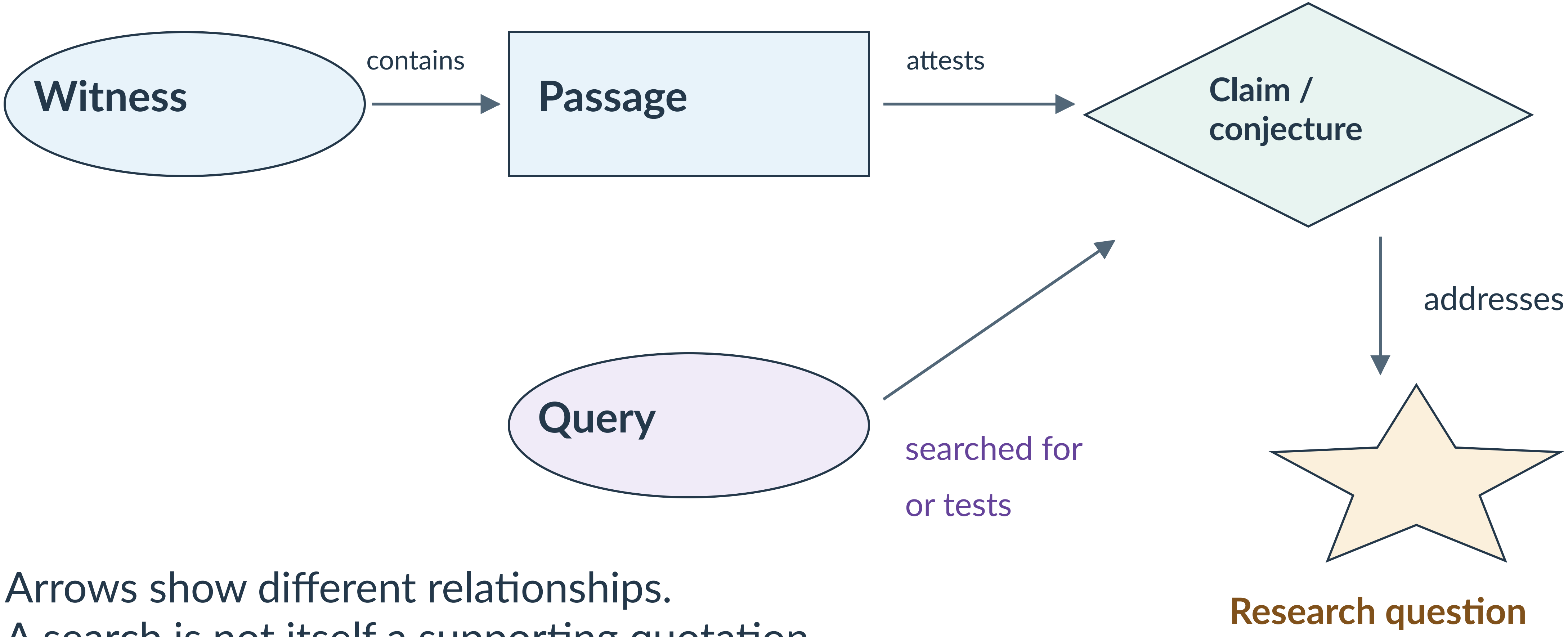
↓

Accept or reject a record

Filtering the view does not delete records.

Screenshot: source passages hidden where applicable. Diagram: schematic.

Why a query points to a proposal



Arrows show different relationships.
A search is not itself a supporting quotation.

Why use an evidence graph?

Trace a proposal	Follow its citations to passages and source records.
Inspect dependencies	See recorded quotation, parallel and descent links.
Keep the history	Who proposed it, what was checked, what the researcher decided.

Future work: edge prediction

Suggest a quotation or parallel to investigate, with passages to check.

Cohort

1	**Find**	Lotus passages and shared wording
2	**Compare**	T0603 with and without its commentary
3	**Inspect**	Saved actions, proposal, citations and checks

Results and limits

Useful outputs

Lotus: a related passage with little exact shared wording.

T0603: exclusion exposes the commentary’s effect on the ranking.

Proposals, sources and checks can be inspected together.

Observed limits

Kumārajīva’s Lotus wrongly ranks Dharmakṣema first in vocabulary.

One known-answer example cannot validate attribution.

Weak tests and reviewer errors still require researcher scrutiny.